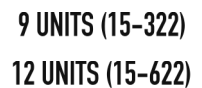


**NOW
FULLY IN
PYTHON**



— continuous signal ○ samples ■ quantized values

$$\text{FM: } y(t) = \sin(2\pi f_c t + I \sin(2\pi f_m t))$$

33-TAP WINDOWED-SINC LOW-PASS FIR

$y[n] = (x * h)[n]$

$|H(f)|$

1

0

0

1 x Nyquist

$$y_L[n] = x[n] + 0.55x[n-D] + 0.25x[n-2D] \quad y_R[n] = x[n] + 0.20x[n-D] + 0.35x[n-3D]$$

columns = next state

rows = current

	C4	E4	G4
C4	0.10	0.60	0.30
E4	0.30	0.20	0.50
G4	0.60	0.30	0.10

→

C C C E C G C C E E G G G C C

16 quarter-note steps



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